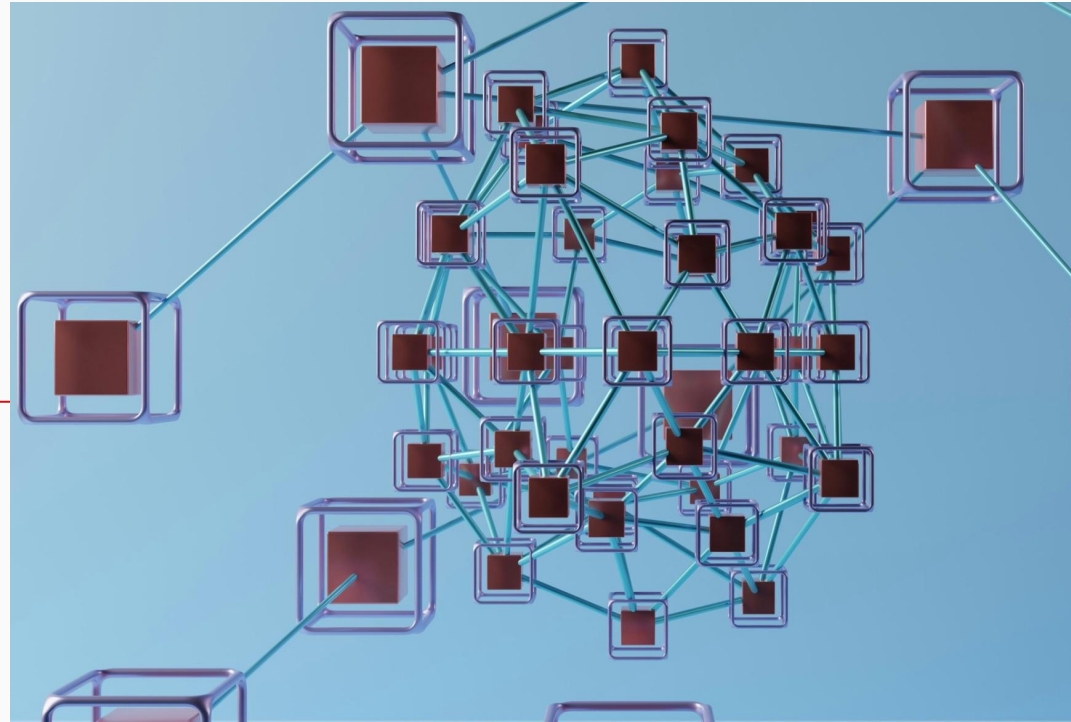


Graph Neural Networks

Aakriti Dhakal



Introduction

Neural Networks

Powerful models that learn patterns from structured data, like images, or text

Limitation - they struggle with irregular or non-grid data

Graphs

Collection of nodes, representing entities, and edges, representing relationships

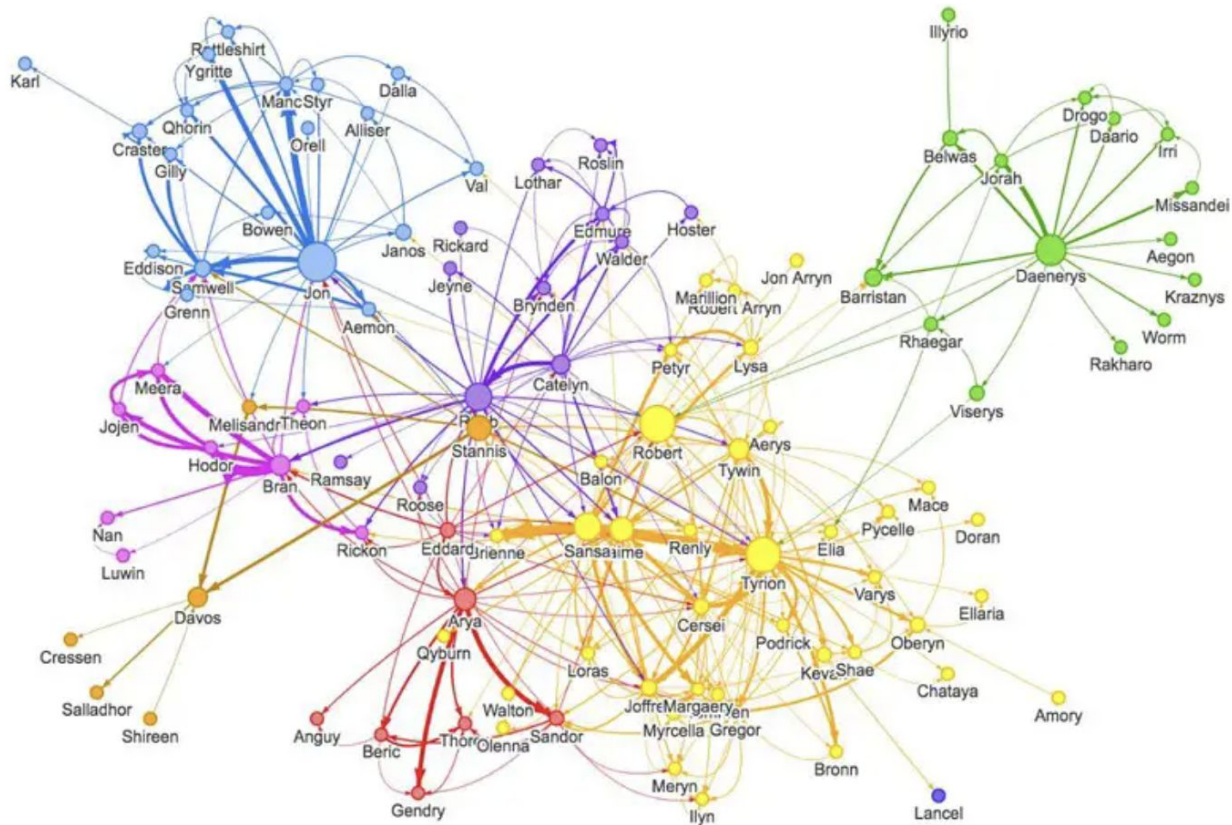
E.g People and friendships in a social network; atoms and chemical bonds in a molecule

GNN

GNNs enable machine learning tasks that leverage the interconnected nature of data

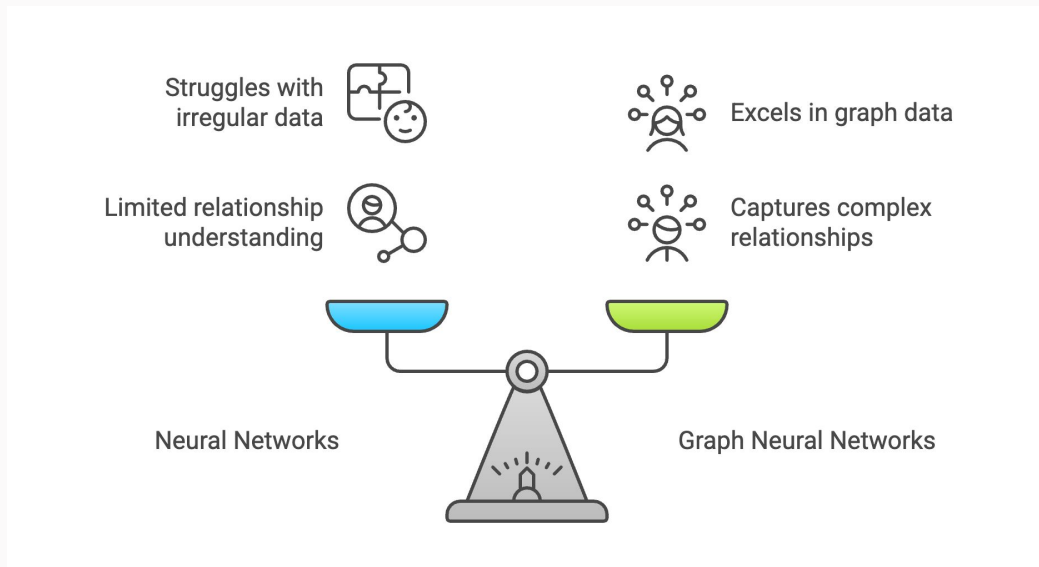
Can learn from both the features of the nodes and the connections between them

Introduction

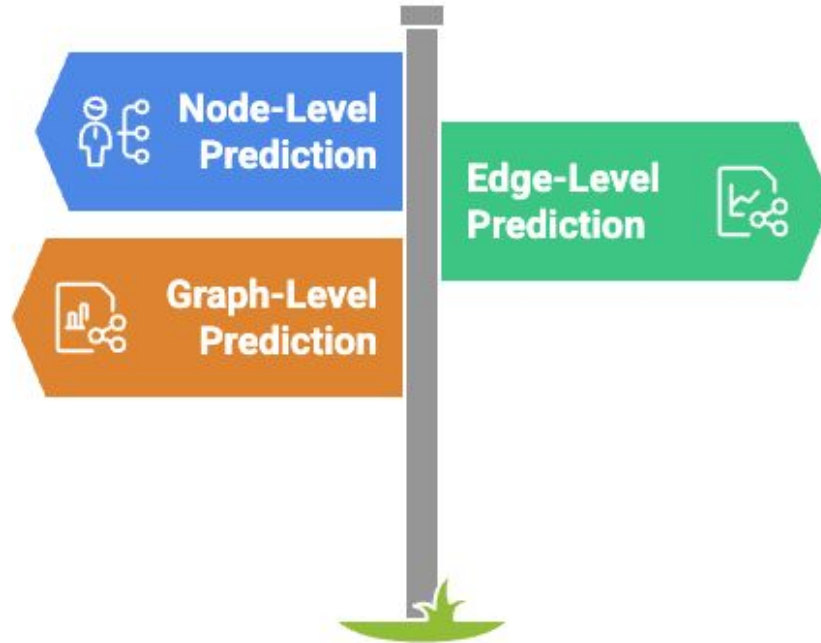


Overview of GNN

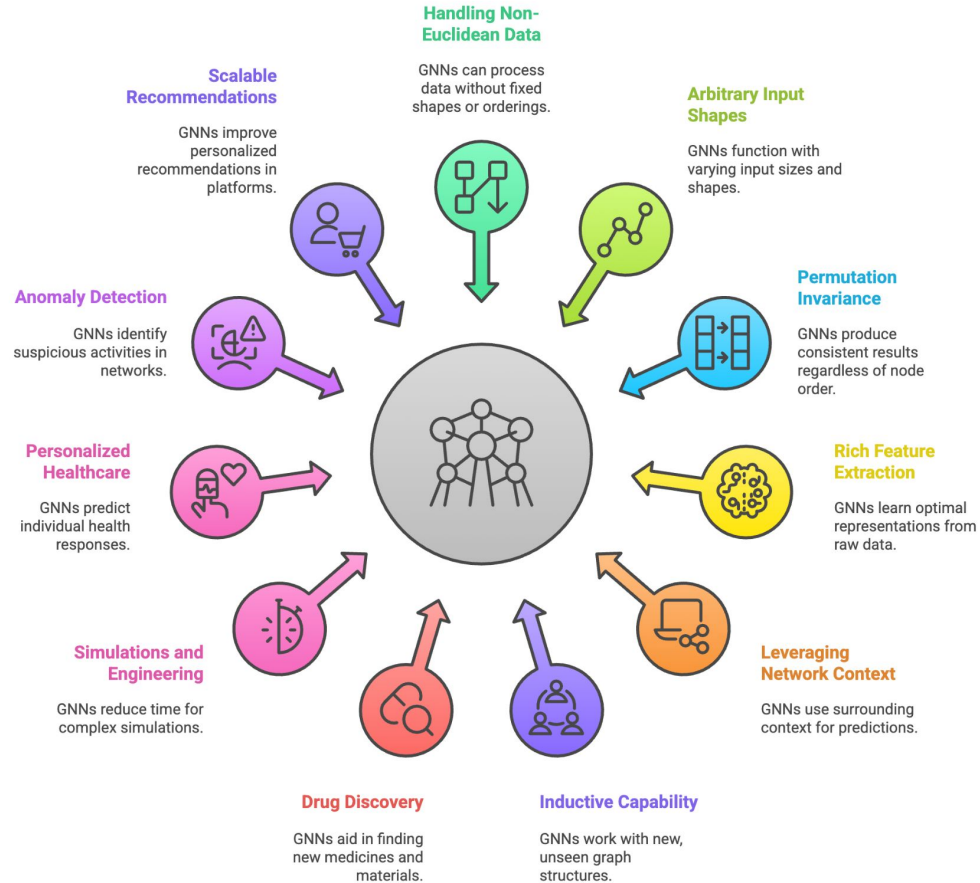
- Deep learning models designed specifically for graphs
- Capture complex relationships & dependencies
- Learn directly from raw graphs
- Allows us to bring the power of neural networks to graph-structured data



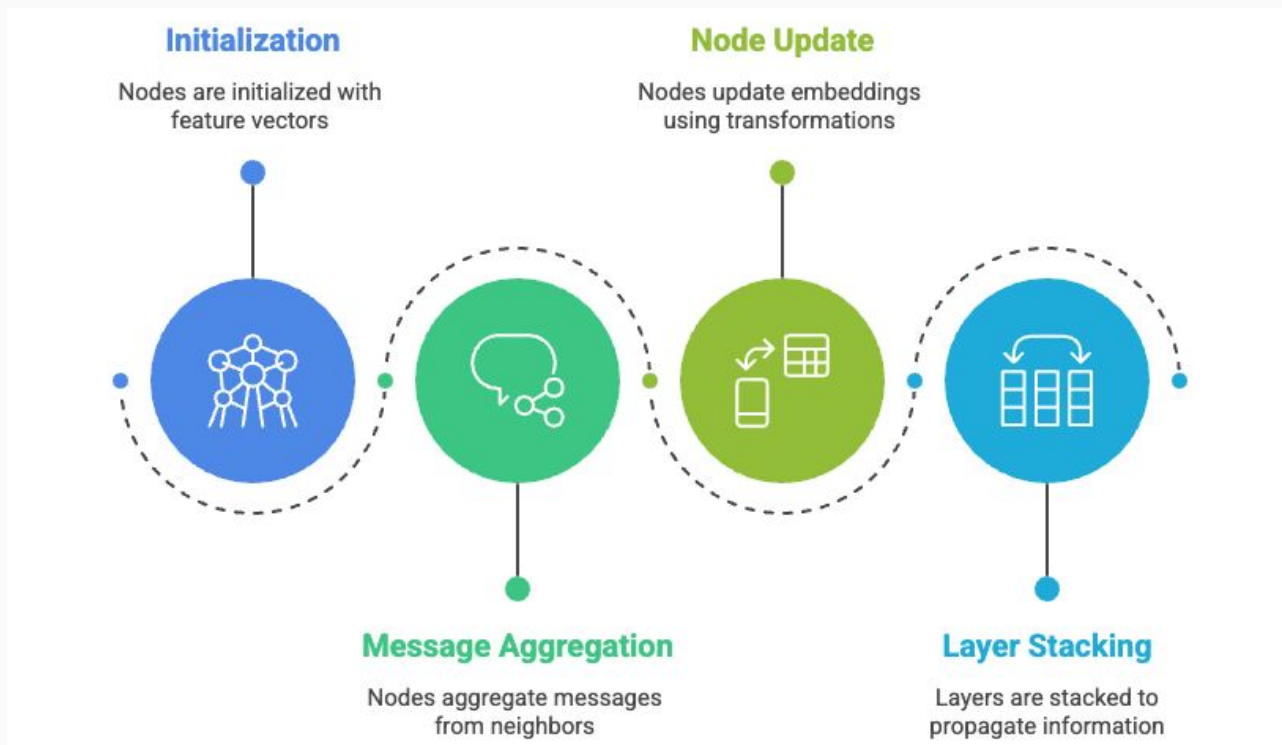
TASKS of GNN



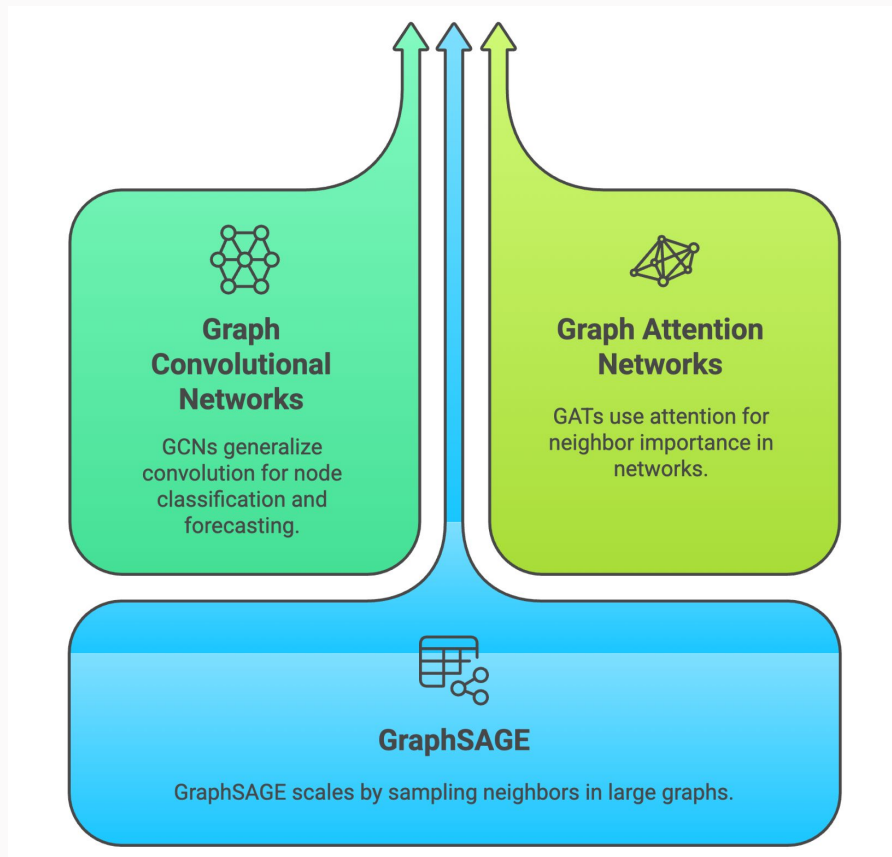
Significance of GNN



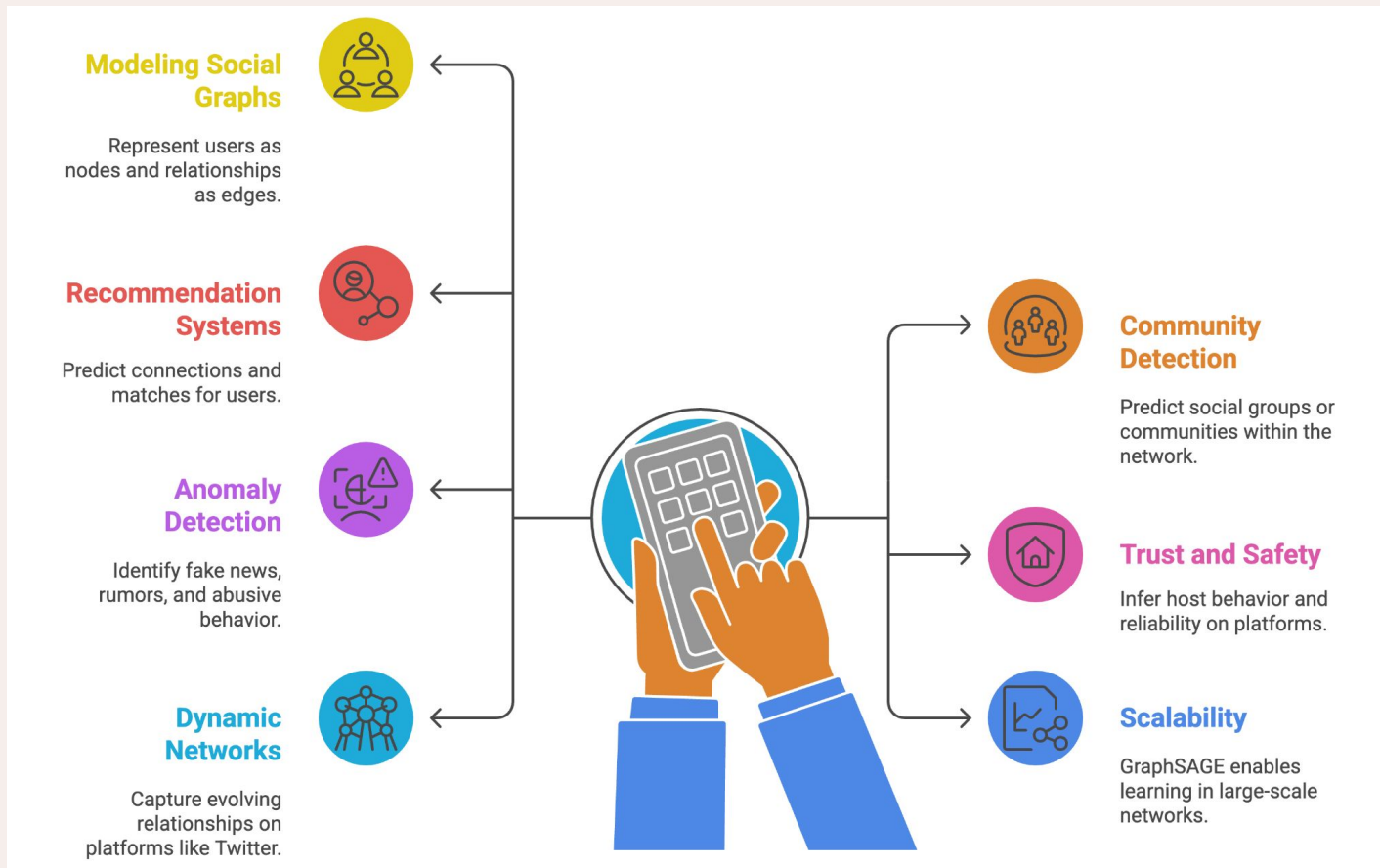
GNN Architecture



GNN Architecture



Use Case : Social Network Analysis



Use Case : *Pinterest*

Pinterest graph:

- 2B pins, 1B boards, 18B+ edges
- Rich ecosystem of visual connections

PinSage (GNN-powered): Scaled recommendation system on Pinterest graph

Key metrics:

- *Hit-rate* : Measures how likely the recommendations include relevant items
- *MRR* : Measures how close relevant items appear to the top of the results

Results:

- +150% Hit-rate
- +60% MRR

Use Case : Molecular Biology

Molecular Representation

Uses nodes and edges to model atoms and bonds.



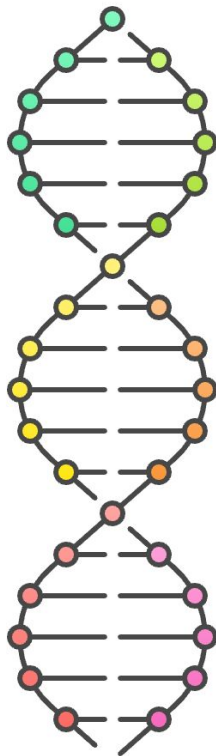
Antibiotic Discovery

Identifies new antibiotics like Halicin using GNNs.



Genomic Analysis

Models gene relationships for prediction tasks.



Drug Discovery & Material Science

Predicts molecular properties and interactions.



Protein-Protein Interaction

Maps biological networks for drug development.



Personalized Medicine

Predicts chemotherapy response using patient data.



Use Case : Protein Design

Protein design: Create new proteins for drugs, enzymes, materials

RFDiffusion (Baker Lab, 2022):

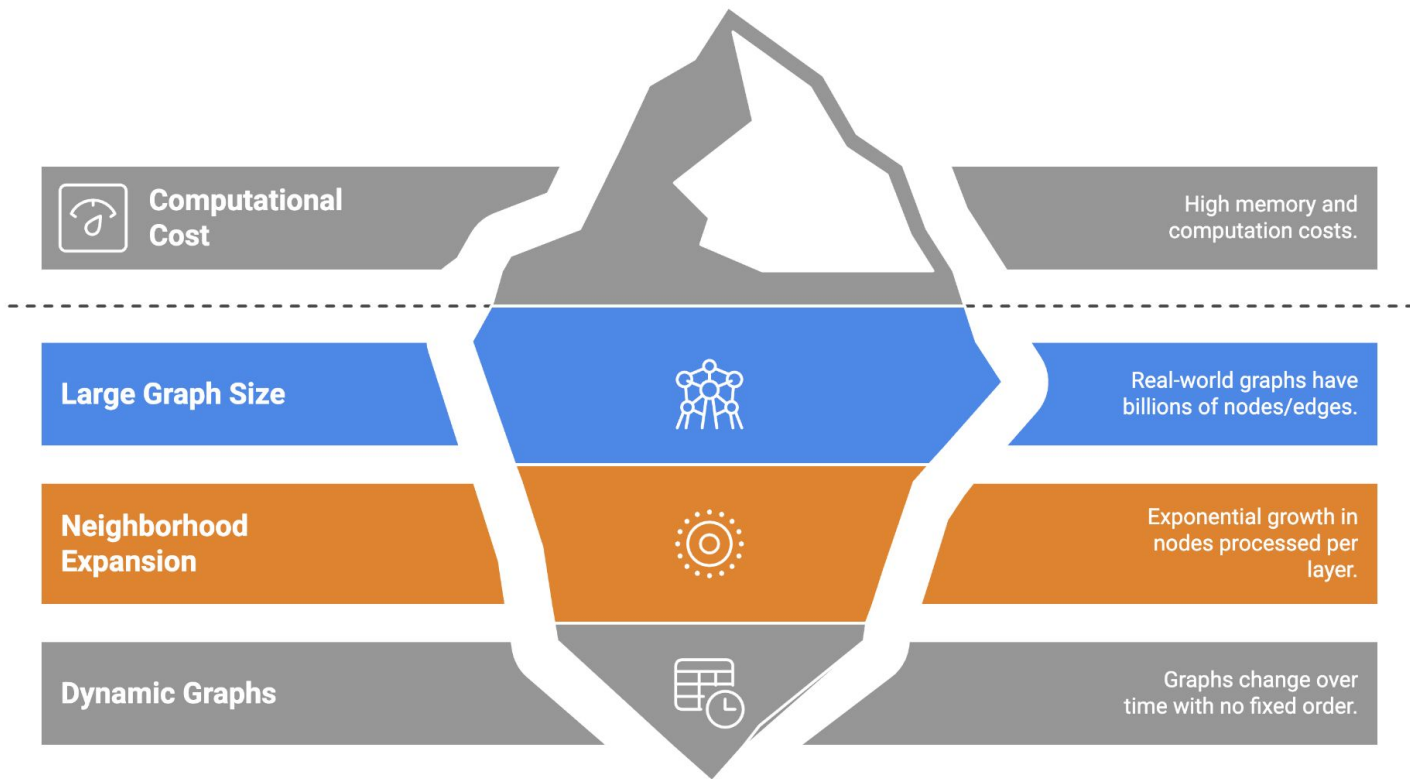
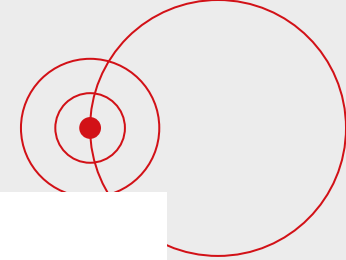
- AI system combining GNN + diffusion models
- It can design protein structures that meet custom constraints

Breakthrough results:

- Solved nearly all benchmark problems (23/25)
- Achieved 18% success rate in designing protein binders - a problems referred to as a “grand challenge” in protein design
- Experimental success rate ranges from 5x to 214× improvement than past methods

Impact: Biggest advance in structural biology, together with AlphaFold

Challenges to address

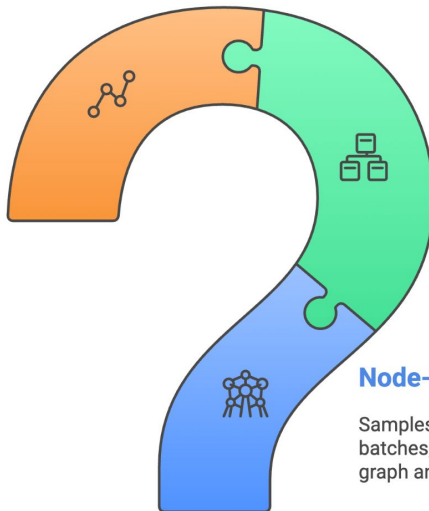


Solutions / Sampling Strategies

Which sampling strategy to use for graph neural network training?

Graph-wise Sampling

Samples subgraphs for mini-batch training, effective for diverse graph structures.



Layer-wise Sampling

Samples nodes per layer, reduces computation, ideal for large graphs.

Node-wise Sampling

Samples neighbors for mini-batches, suitable for local graph analysis.

Future Directions in GNN Research



Graph Sparsification

Reduce edges while preserving critical information



Graph Coarsening

Simplify graphs hierarchically for faster processing



Parallel Computing

Leverage GPUs and TPUs for large-scale datasets



Sampling Trade-offs

Balance variance and bias in neighbor sampling



Complex Graph Types

Design models for heterogeneous and dynamic graphs

GNN in the AI Research Landscape

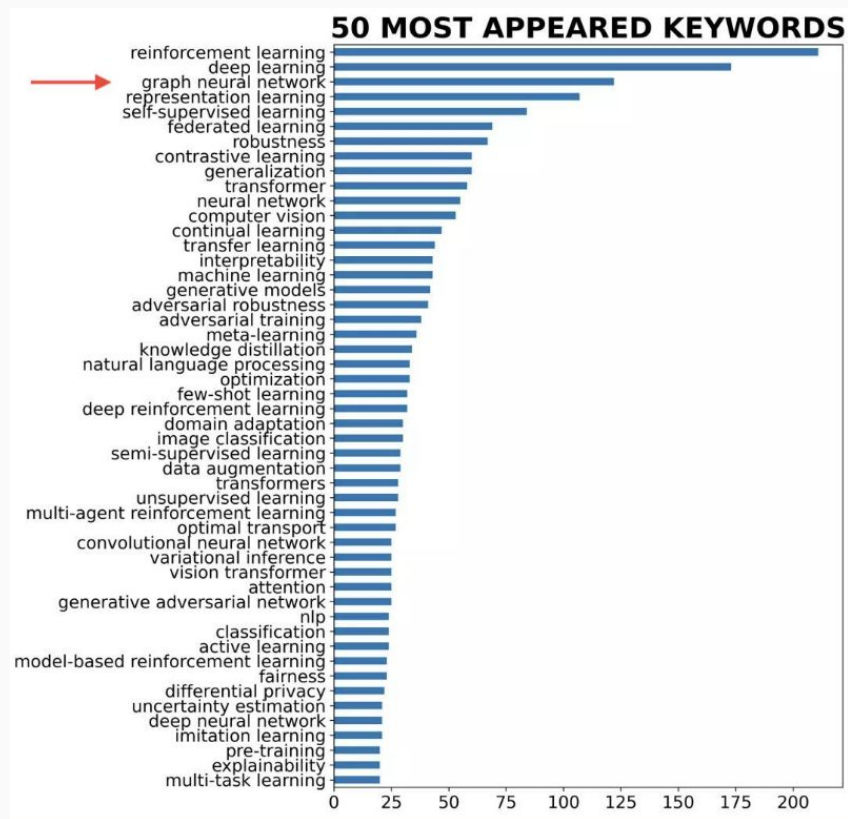


Fig: Top 50 keywords in submitted research papers at ICLR 2022

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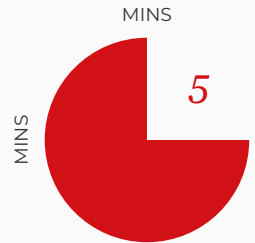
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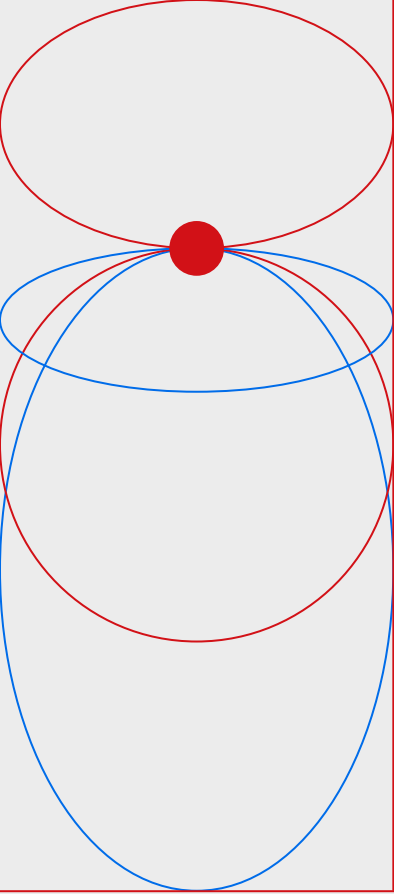
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Open discussion

Any questions/review
about the
presentation?





*Thank
you*